

# HDOC DAY 38

## Problem

Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

## C code

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>

void length()

{

    char sen[100];

    printf("enter a sentence\n");

    fgets(sen,sizeof(sen),stdin);

    sen[strcspn(sen,"\n")]='\0';

    int spaces=0,digits=0,special=0;

    int len=strlen(sen);

    for(int i=0;i<len;i++)

    {

        char c=sen[i];

        if(c==' ')

        {
```

```

        spaces++;
    }
    else if(isdigit(c))
    {
        digits++;
    }
    else if(!isalnum(c))
    {
        special++;
    }
}

printf("the length of the string is %d\n",len);
printf("the number of spaces is %d\n",spaces);
printf("the number of digits is %d\n",digits);
printf("the number of special characters is %d\n",special);
}

void print()
{
    char sen[100];
    char sen1[100];
    char chh;
    int freq=0;
    printf("enter a sentence\n");
    fgets(sen, sizeof(sen), stdin);
    while (getchar() != '\n');
    int len=strlen(sen);
    printf("enter the charecter to be searched in a string\n");
    scanf("%c",&chh);

```

```

        for(int i=0;i<len;i++)
        {
            char c=sen[i];
            if(c==chh)
            {
                freq++;
            }
        }
        sen1[len]='\0';
        printf("the frequency of the character is %d\n",freq);
    }
    void vowcon()
    {
        char sen[100];
        printf("enter a sentence\n");
        fgets(sen, sizeof(sen), stdin);
        sen[strcspn(sen, "\n")] = '\0';
        int len=strlen(sen);
        for(int i=0;i<len;i++)
        {
            if(sen[i]!=' ')
            {
                if(islower((unsigned char)sen[i]))
                {
                    sen[i]=toupper((unsigned char)sen[i]);
                }
            }
            else
            {

```

```

        sen[i]=tolower((unsigned char)sen[i]);
    }
}
}
printf("the string after changing the case is \n%s",sen);
}
int main()
{
    int ch;
    printf("choice 1 is to find the length of a string(not using a in built function)\n");
    printf("choice 2 is to print each character of a string in a new line\n");
    printf("choice 3 is to Count vowels and consonants in a string.\n");
    printf("exit is choice 4\n");
    printf("enter your choice\n");
    scanf("%d",&ch);
    while (getchar() != '\n');
    switch (ch)
    {
        case 1:
        {
            length();
            break;
        }
        case 2:
        {
            print();
            break;
        }
    }
}

```

```
case 3:
{
    vowcon();
    return 0;
}
case 4:
{
    printf("exit\n");
    return 0;
}
default:
{
    printf("invalid choice\n");
    break;
}
}
return 0;
}
```